Year 13 Further Maths - Powers and roots of complex numbers

Similar questions - nth roots in exponential form

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

(a) Determine the square roots of 36i in the form $re^{i\theta}$, where $0 \leqslant \theta < 2\pi$. **[3]**

(b) Illustrate 36i and its square roots on an Argand diagram. **[1]**

2 In this question you must show detailed reasoning. The complex number $-4\sqrt{2}+4\sqrt{2}i$ is denoted by z.

(a) Determine the cube roots of z, giving the roots in exponential form. **[6]**

(b) The cube roots are represented by A, B, C, forming a triangle. Write down the angles that any lines of symmetry of the triangle make with the positive real axis, justifying your answer. **[3]**

3

(a) Show that $\dfrac{-1+i\sqrt{3}}{2} = e^{\frac{2}{3}\pi i}$. **[2]**

(b) Hence determine the exact roots of $z^4 = 8\left(-1+i\sqrt{3}\right)$, giving them in the form $re^{i\theta}$ where $r > 0$ and $0 \leqslant \theta < 2\pi$. **[4]**

Answers

Question 1

(a) $36\mathrm{i} = 36\mathrm{e}^{\mathrm{i}\pi/2}$, so $z = 6\mathrm{e}^{\mathrm{i}(\pi/4+k\pi)}$: $z = 6\mathrm{e}^{\mathrm{i}\pi/4}$ and $z = 6\mathrm{e}^{\mathrm{i}5\pi/4}$

(b) 36i is on the positive imaginary axis at distance 36; the roots are at distance 6 from the origin on the line $y = x$, at $\frac{\pi}{4}$ and $\frac{5\pi}{4}$ (diametrically opposite)

Question 2

(a) $|z| = \sqrt{32+32} = 8$, $\arg z = \frac{3\pi}{4}$. $z = 8\mathrm{e}^{\mathrm{i}(3\pi/4+2k\pi)}$. Cube roots: $2\mathrm{e}^{\mathrm{i}(\pi/4+2k\pi/3)}$: $2\mathrm{e}^{\mathrm{i}\pi/4}$, $2\mathrm{e}^{\mathrm{i}11\pi/12}$, $2\mathrm{e}^{\mathrm{i}19\pi/12}$

(b) The triangle is equilateral (vertices equally spaced on a circle, centre O), so each line of symmetry passes through O and one vertex: angles $\frac{\pi}{4}$, $\frac{11\pi}{12}$ and $\frac{19\pi}{12}$ with the positive real axis.

Question 3

(a) $\cos\frac{2\pi}{3} = -\frac{1}{2}$ and $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

(b) $8(-1+\mathrm{i}\sqrt{3}) = 16\mathrm{e}^{\frac{2}{3}\pi\mathrm{i}}$. Fourth roots: $2\mathrm{e}^{\mathrm{i}(\frac{\pi}{6}+\frac{k\pi}{2})}$: $2\mathrm{e}^{\mathrm{i}\pi/6}$, $2\mathrm{e}^{\mathrm{i}2\pi/3}$, $2\mathrm{e}^{\mathrm{i}7\pi/6}$, $2\mathrm{e}^{\mathrm{i}5\pi/3}$